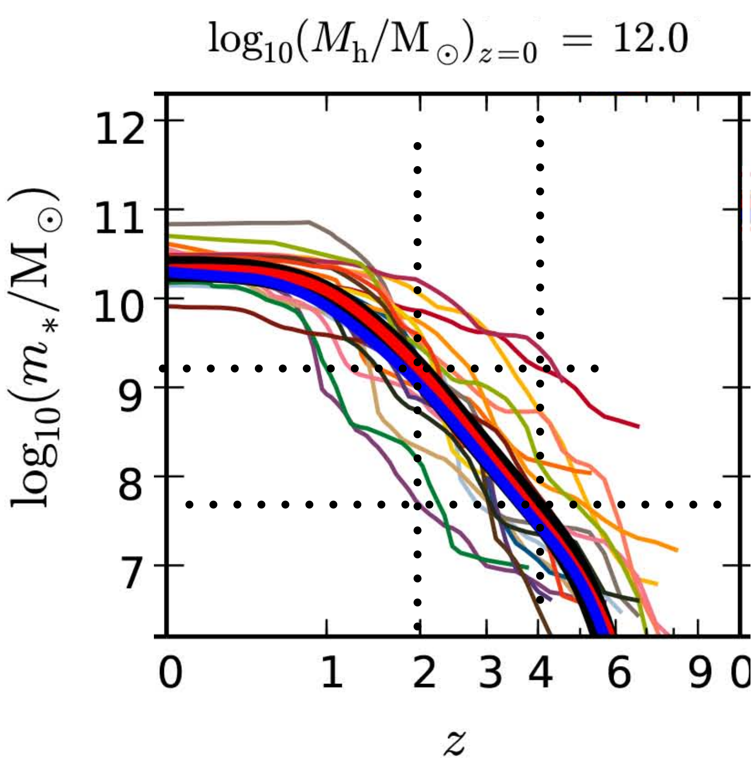
Expected Stellar Mass for AGORA halo at z=4 is (0.5-1)x10^9 $M_\odot$ at z=2 is (4-8)x10^9 $M_\odot$

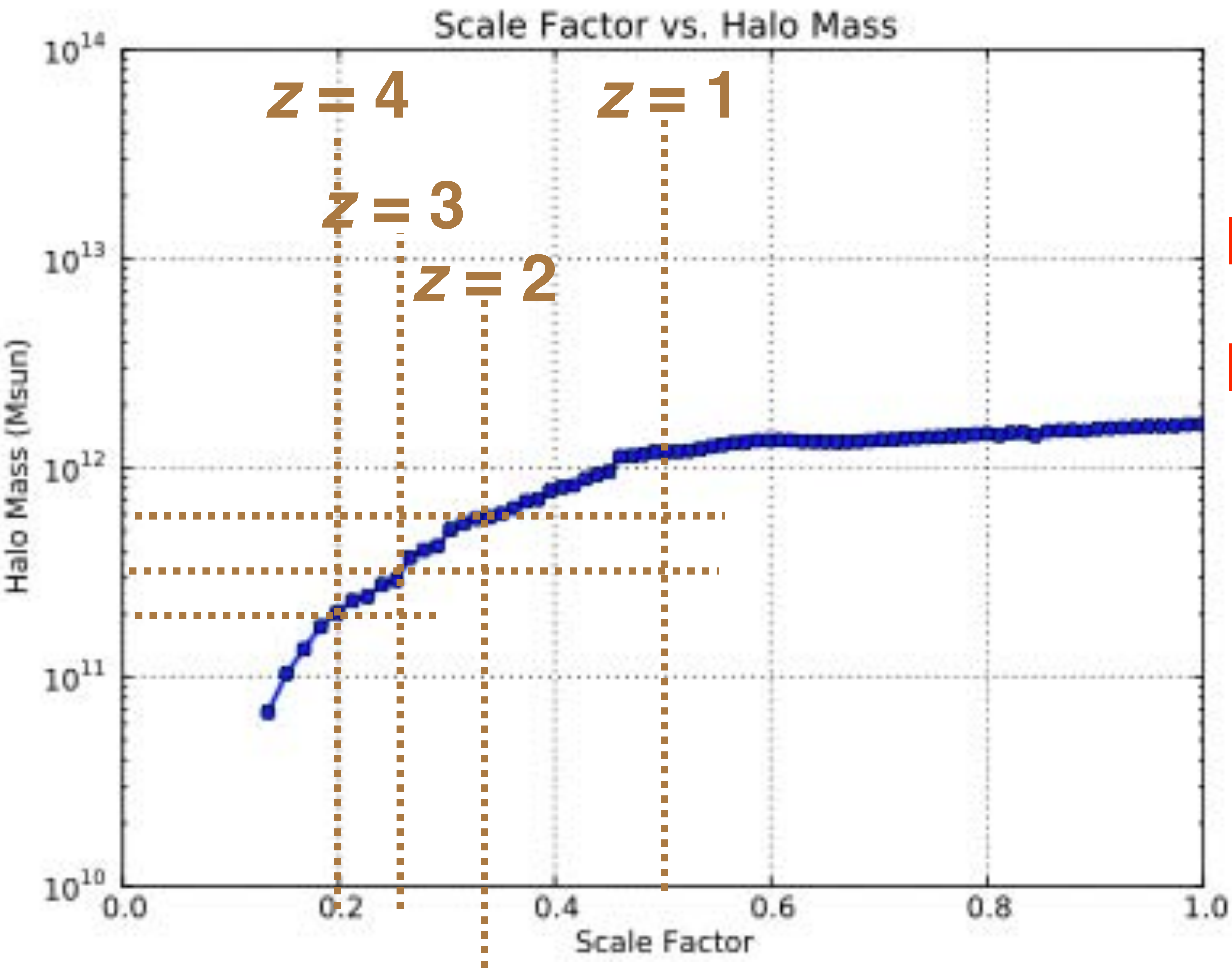
But Moster+2018 shows that the dispersion in 20 random halos is wider than the factor of 2 above (see next to last page below, copied at right)



$M_{vir} = 0.5 \times 10^{11}$ at $z = 4$ for typical $10^{12}\ M_\odot$ halo at $z = 0$

$M_{vir} = 1 \times 10^{11}$ at $z = 4$ for typical $1.7 \times 10^{12}\ M_\odot$ halo at $z = 0$

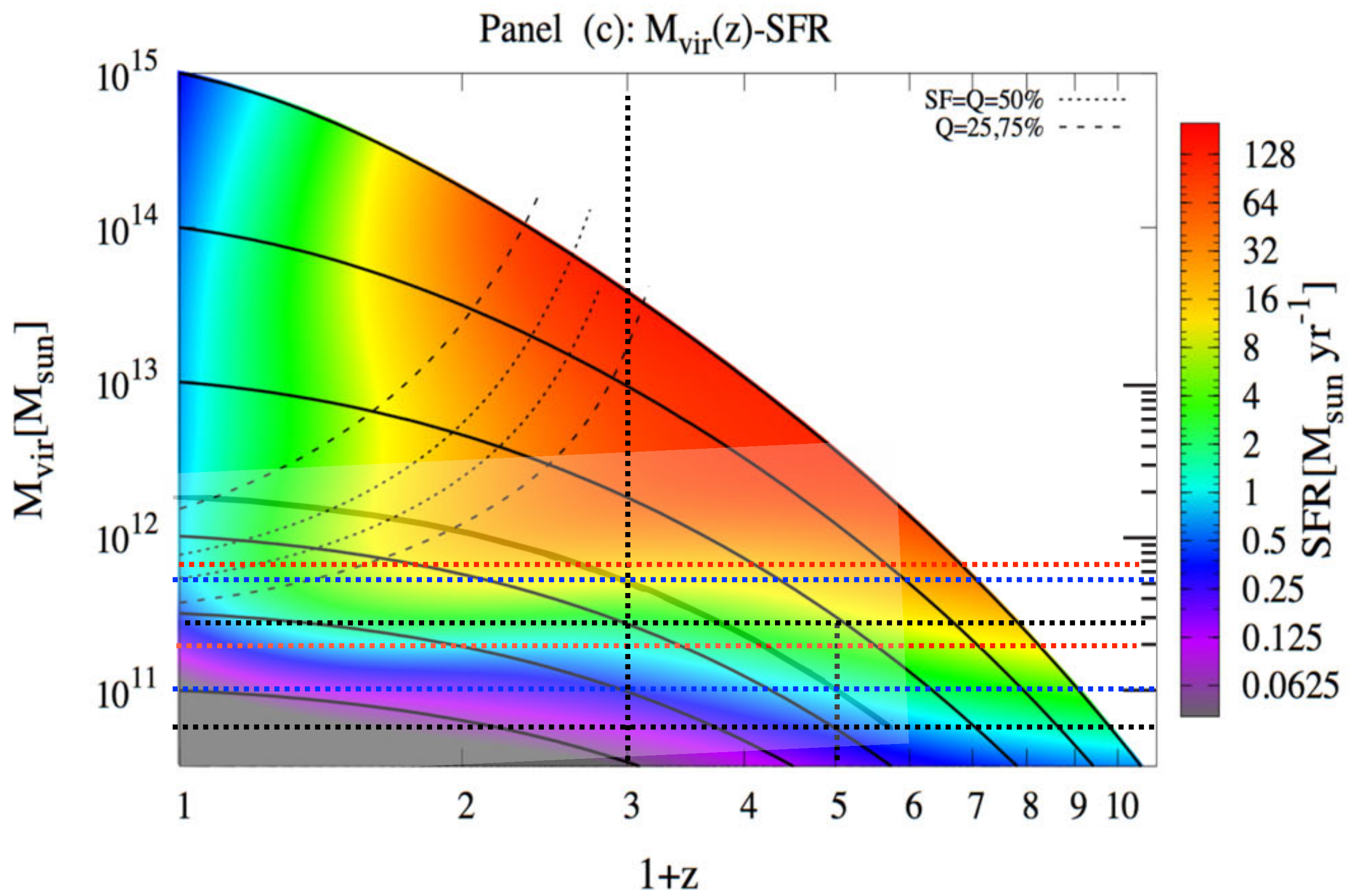
$M_{vir} = 2 \times 10^{11}$ at $z = 4$ for AGORA $1.7 \times 10^{12}\ M_\odot$ halo at $z = 0$



AGORA halo

$M_{vir} = 3 \times 10^{11}$ at $z = 3$

$M_{vir} = 6 \times 10^{11}$ at $z = 2$



Constraining the galaxy–halo connection over the last 13.3 Gyr: star formation histories, galaxy mergers and structural properties

Aldo Rodŕıguez-Puebla, Joel R. Primack, Vladimir Avila-Reese, S. M. Faber 2017

$M_{vir} = 2\times10^{11}$ at $z = 4$ for AGORA 1.7×10^{12} $M_{\odot}$ halo at $z = 0$

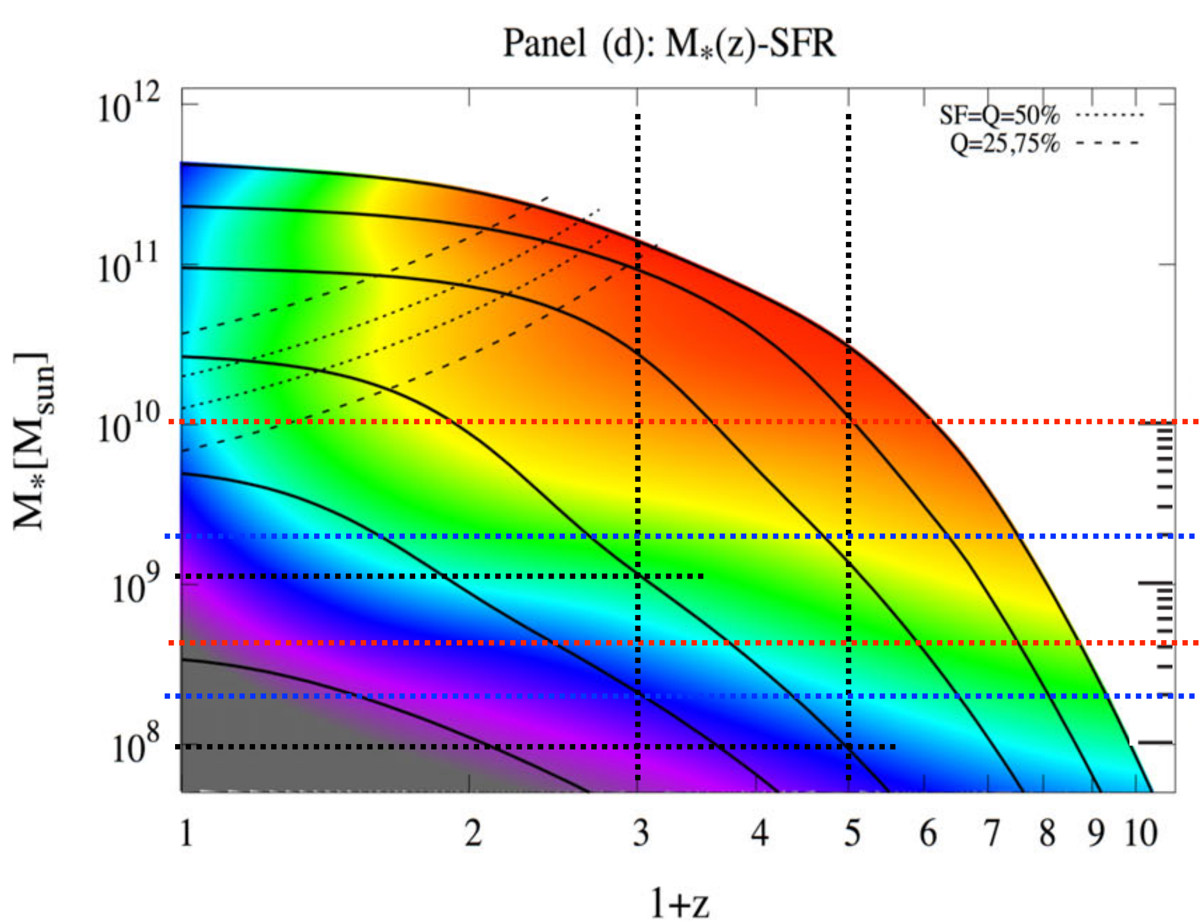
$M_{vir} = 1\times10^{11}$ at $z = 4$ for typical 1.7×10^{12} $M_{\odot}$ halo at $z = 0$

$M_{vir} = 0.6\times10^{11}$ at $z = 4$ for typical 10^{12} $M_{\odot}$ halo at $z = 0$

$M_{vir} = 6\times10^{11}$ at $z = 2$ for AGORA 1.7×10^{12} $M_{\odot}$ halo at $z = 0$

$M_{vir} = 5\times10^{11}$ at $z = 2$ for typical 1.7×10^{12} $M_{\odot}$ halo at $z = 0$

$M_{vir} = 3\times10^{11}$ at $z = 2$ for typical 10^{12} $M_{\odot}$ halo at $z = 0$



$M^*/M_{\odot}$ at z = 4

	Rodriguez-Puebla+17	UnivMachine2018
typical10^{12} halo at z=0	1×10^{8}	1×10^{8}
typical 1.7×10^{12} halo	2×10^{8}	3×10^{8}
AGORA 1.7×10^{12} halo	4×10^{8}	1×10^{9}

$M^*/M_{\odot}$ at z = 2

	Rodriguez-Puebla+17	UnivMachine2018
typical10^{12} halo at z=0	1.5×10^{9}	2×10^{9}
typical 1.7×10^{12} halo	4×10^{9}	5×10^{9}
AGORA 1.7×10^{12} halo	5×10^{9}	7×10^{9}

Constraining the galaxy–halo connection over the last 13.3 Gyr: star formation histories, galaxy mergers and structural properties

Aldo Rodr´ıguez-Puebla, Joel R. Primack, Vladimir Avila-Reese, S. M. Faber 2017

M_{vir} = 2x10^11 at $z = 4$ for AGORA 1.7x10^12 $M_\odot$ halo at $z = 0$

M_{vir} = 1x10^11 at $z = 4$ for typical 1.7x10^12 $M_\odot$ halo at $z = 0$

M_{vir} = 0.6x10^11 at $z = 4$ for typical 10^12 $M_\odot$ halo at $z = 0$

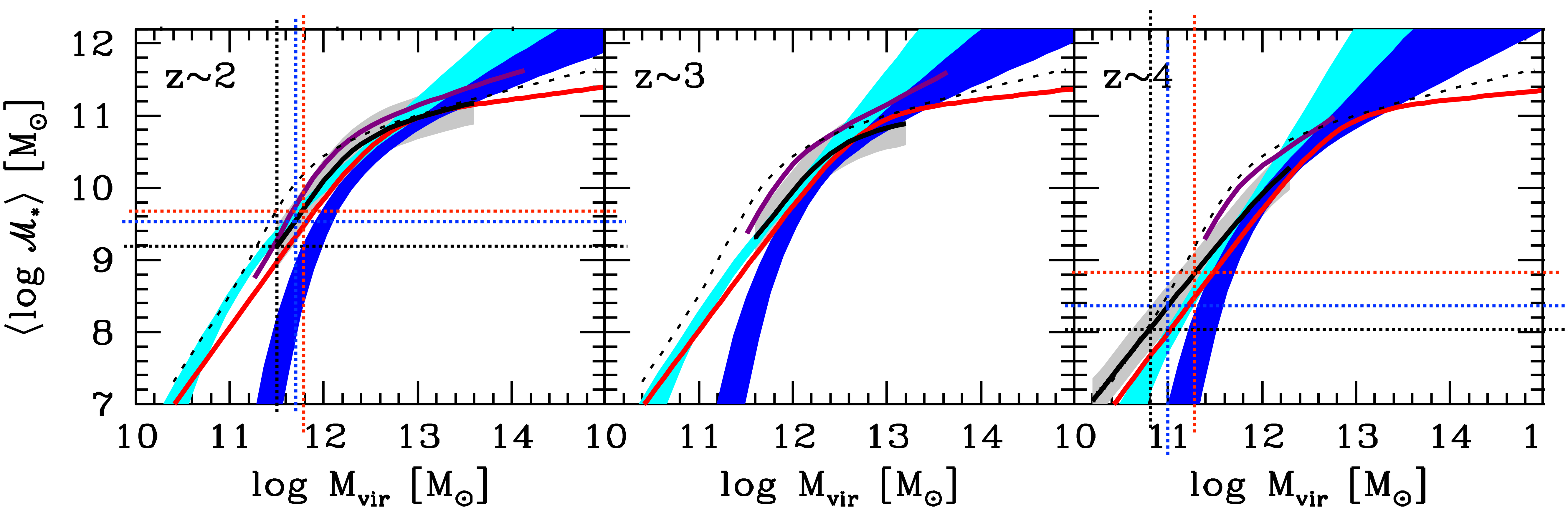
M_{vir} = 6x10^11 at $z = 2$ for AGORA 1.7x10^12 $M_\odot$ halo at $z = 0$

M_{vir} = 5x10^11 at $z = 2$ for typical 1.7x10^12 $M_\odot$ halo at $z = 0$

M_{vir} = 3x10^11 at $z = 2$ for typical 10^12 $M_\odot$ halo at $z = 0$

	$M^*/M_\odot$ at z = 2		$M^*/M_\odot$ at z = 4	
	Rodriguez-Puebla+17	UnivMachine2018	Rodriguez-Puebla+17	UnivMachine2018
typical10^12 halo at z=0	1.5x10^9	2x10^9	1x10^8	1x10^8
typical 1.7x10^12 halo	4x10^9	5x10^9	2x10^8	3x10^8
AGORA 1.7x10^12 halo	5x10^9	7x10^9	6x10^8	1x10^9

This work
Guo+2010
Yang+2012 (SMF1)
Yang+2012 (SMF2)
Behroozi+2013
Moster+2013



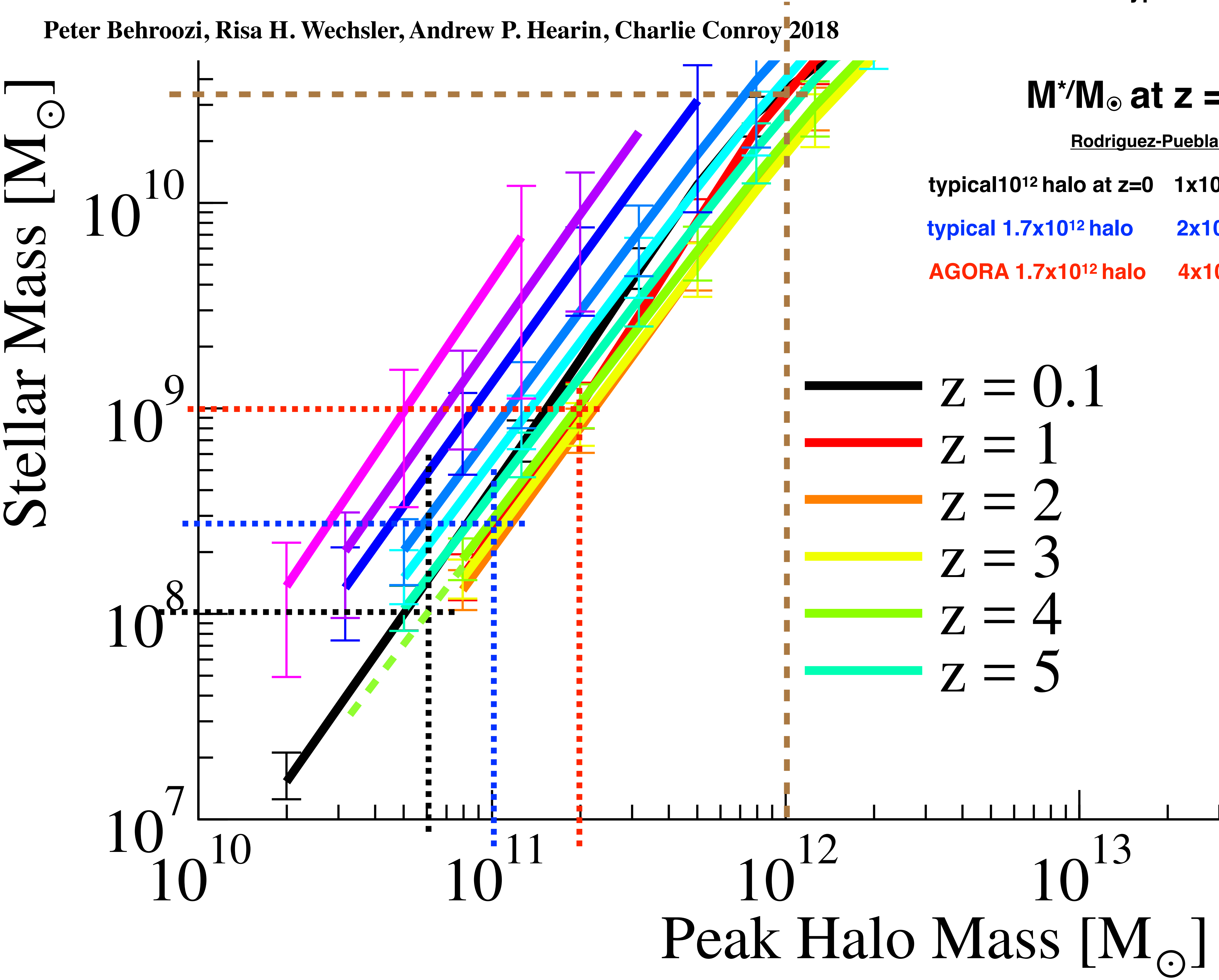
UNIVERSEMACHINE: The Correlation between Galaxy Growth and Dark Matter Halo Assembly from z = 0-10

Peter Behroozi, Risa H. Wechsler, Andrew P. Hearin, Charlie Conroy 2018

$M_{vir} = 2\times10^{11}$ at $z = 4$ for AGORA 1.7×10^{12} $M_\odot$ halo at $z = 0$

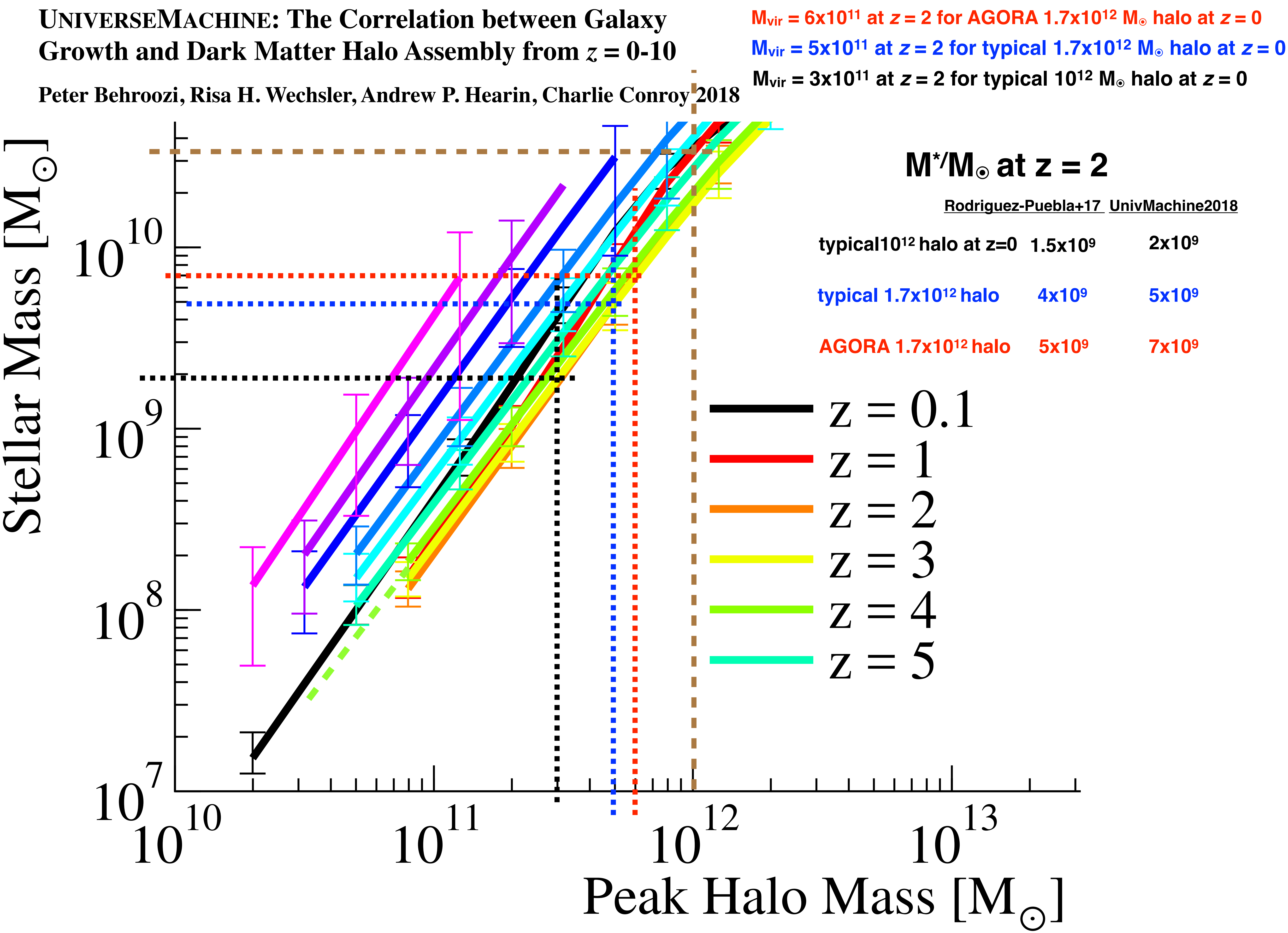
$M_{vir} = 1\times10^{11}$ at $z = 4$ for typical 1.7×10^{12} $M_\odot$ halo at $z = 0$

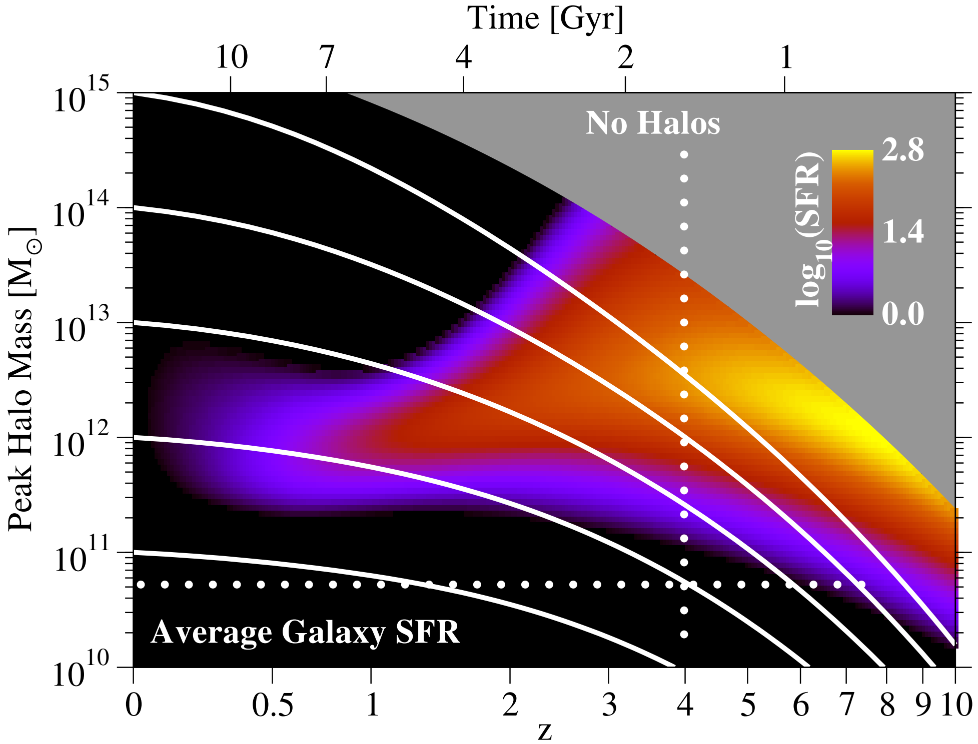
$M_{vir} = 0.6\times10^{11}$ at $z = 4$ for typical 10^{12} $M_\odot$ halo at $z = 0$



$M^*/M_\odot$ at z = 4

	Rodriguez-Puebla+17	UnivMachine2018
typical 10^{12} halo at z=0	1×10^8	1×10^8
typical 1.7×10^{12} halo	2×10^8	3×10^8
AGORA 1.7×10^{12} halo	4×10^8	1×10^9


UNIVERSEMACHINE: The Correlation between Galaxy Growth and Dark Matter Halo Assembly from z = 0-10
Peter Behroozi, Risa H. Wechsler, Andrew P. Hearin, Charlie Conroy 2018
M_{vir} = 6x10^{11} at z = 2 for AGORA 1.7x10^{12} $M_\odot$ halo at z = 0
M_{vir} = 5x10^{11} at z = 2 for typical 1.7x10^{12} $M_\odot$ halo at z = 0
M_{vir} = 3x10^{11} at z = 2 for typical 10^{12} $M_\odot$ halo at z = 0
$M^*/M_\odot$ at z = 2
Rodriguez-Puebla+17 UnivMachine2018
typical10^{12} halo at z=0 1.5x10^9 2x10^9
typical 1.7x10^{12} halo 4x10^9 5x10^9
AGORA 1.7x10^{12} halo 5x10^9 7x10^9
z = 0.1
z = 1
z = 2
z = 3
z = 4
z = 5
Stellar Mass [$M_\odot$]
10^7
10^8
10^9
10^{10}
Peak Halo Mass [$M_\odot$]
10^{10}
10^{11}
10^{12}
10^{13}



UNIVERSEMACHINE: The Correlation between Galaxy Growth and Dark Matter Halo Assembly from z = 0-10

Peter Behroozi, Risa H. Wechsler, Andrew P. Hearin, Charlie Conroy 2018

Typical 10^{12} halo at z=0 has $M_h = 5\mathrm{x}10^{10}$ at z=4 with $M^*/M_h= 0.002 \implies M^*=10^8$ at z=4.

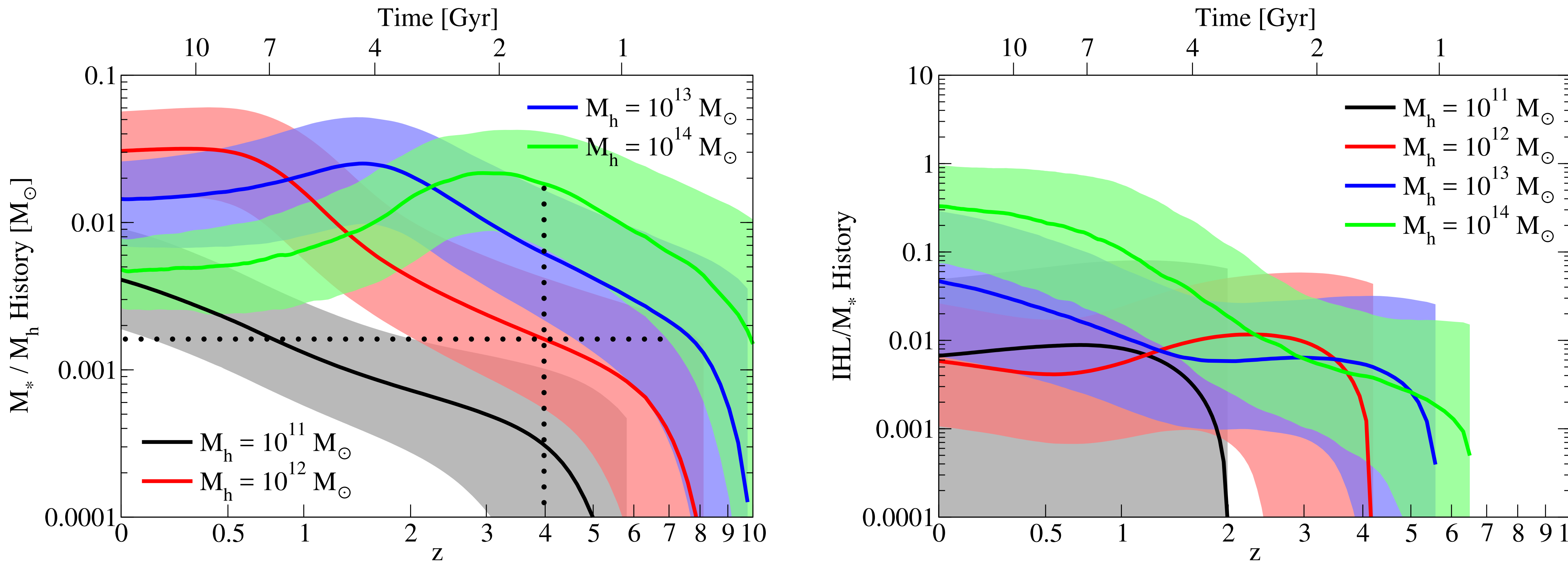


Figure 16. Top left: total (including mergers) star formation histories for haloes in bins of M_{peak} for our best-fitting model; colored bars indicate the $16^{\mathrm{th}} - 84^{\mathrm{th}}$ percentile range among different haloes in the best-fitting model. **Top right**: same, with main progenitor stellar mass histories. **Middle left**: same, with main progenitor SSFR histories. **Middle right**: same, with main progenitor intrahalo light (IHL) histories. **Bottom left**: same, with main progenitor M_*/M_{peak} ratio histories. **Bottom right**: same, with main progenitor IHL / M_* ratio histories. For all panels, bin widths are ±0.25 dex; e.g., the label $M_h = 10^{11}\ \mathrm{M}_\odot$ corresponds to $10.75 < \log_{10}(M_{\mathrm{peak}}/\ \mathrm{M}_\odot) < 11.25$. **Notes:** Quiescent SSFRs in our models are fixed to $10^{-11.8}\ \mathrm{yr}^{-1}$ (§3.2), explaining why massive haloes' SSFRs are close to this value at low redshifts. The feature at low redshift in the 68% confidence contour for SSFRs of $10^{13}\ \mathrm{M}_\odot$ haloes arises because of the same issue as discussed in the notes for Figs. 2 and 11.

EMERGE – an empirical model for the formation of galaxies since $z \sim 10$

Moster Naab White 2018

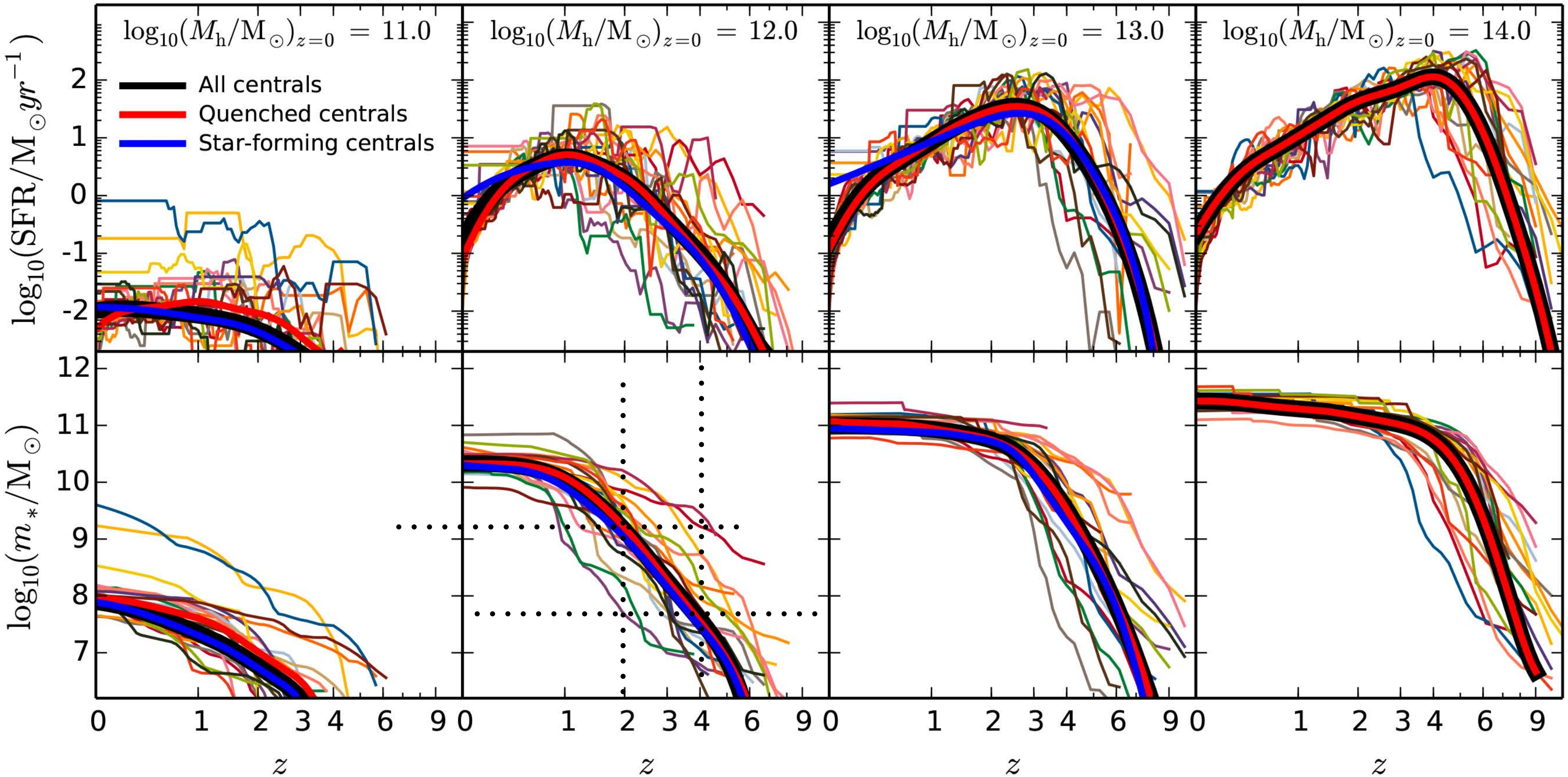


Figure 10. Stellar mass build-up in haloes of different mass. Top panels: SFR as a function of redshift for central galaxies in haloes of $10^{11} - 10^{14}\,\mathrm{M}_\odot$ at $z = 0$ (from left to right). Thin lines are tracks for 20 randomly chosen haloes. Individual tracks can differ significantly in particular at high redshift and for low present-day halo masses. The black, red, and blue thick lines indicate the ensemble averages for all, quenched and star-forming centrals, respectively. For higher halo masses star formation peaks at higher values and higher redshifts. Bottom panels: stellar mass as a function of redshift for the same $z = 0$ halo masses. Lines are the same as in the top panels, i.e. lines of the same colour represent the same halo.

$M^*/M_\odot$ at z = 4

	Rodriguez-Puebla+17	UnivMachine2018
typical 10^{12} halo at z=0	1×10^8	1×10^8
typical 1.7×10^{12} halo	2×10^8	3×10^8
AGORA 1.7×10^{12} halo	4×10^8	1×10^9

$M^*/M_\odot$ at z = 2

	Rodriguez-Puebla+17	UnivMachine2018
typical 10^{12} halo at z=0	1.5×10^9	2×10^9
typical 1.7×10^{12} halo	4×10^9	5×10^9
AGORA 1.7×10^{12} halo	5×10^9	7×10^9